

Phase Synchronization for Distributed Coherent Arrays

From oscillator physics to counted drone detections — a waveform-level simulation study

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Every number in this deck is regenerated from the accompanying code (seed 0); commands to reproduce each figure are on the final slide.

Roadmap

- 1 Motivation: why phase coherence is worth fighting for
- 2 Method: a waveform-level, impairment-complete testbed
- 3 Synchronization results: six methods, one scoreboard
- 4 Scaling to N stations: the airtime wall
- 5 The application: counted drone detections
- 6 The frontier: synchronization as a scheduled resource
- 7 Novelty, future directions, and honest limits

The prize: N radios that act as one aperture

If N base stations share one carrier phase, their fields *add* at a target:

- Transmit focusing: $N^2 \times$ one station's power ($2 \times$ from total power, $2 \times$ again because fields add — 2 stations in phase deliver $4 \times$).
- Radar (round trip): another N on receive,

$$\text{SNR}_{\text{det}} = N^3 G^2 \times \text{single station},$$

where $G \in [0, 1]$ is the *array coherent gain* and enters **squared** — sync errors cost both legs.

No sync at all

Two free-running transmitters average $G = 50\%$ and sweep through 0% (fields cancel) as their phases drift through the circle. The coherent bonus simply does not exist.

The community yardstick

Total phase error $\leq 18^\circ$ (314 mrad) $\Rightarrow \geq 90\%$ of ideal gain at large N ($G_2 = \cos^2(\theta/2)$; at $N = 2$, 18° still gives 97.6%).

The obstacle: every station has its own clock

The 915 MHz carrier is synthesized from an imperfect crystal; two stations disagree in frequency *and* phase:

class	example	accuracy	initial CFO
OCXO	macro BS, GPS-disciplined	± 50 ppb	~ 46 Hz
TCXO	small cell	± 100 ppb	~ 92 Hz
SDR	bench USRP, no GPSDO	± 2 ppm	~ 1830 Hz

Even after frequency lock the phase *random-walks*: white-FM noise alone accumulates ~ 45 mrad RMS per 50 ms at the default class. The SDR class cannot hold carrier phase at all at 50 ms cadence (residual ~ 2 rad).

The measurement problem

A radio can only measure

$$m = \underbrace{\theta}_{\text{oscillators}} + \underbrace{\phi_c}_{\text{channel}}$$

— the two are entangled in every pilot.
Separating them is half the algorithm design.

Standards don't cover this

3GPP sync budgets are *time* alignment (μs); a JSAC 2023 cell-free prototype measured $> 20^\circ$ LO drift per 7.5 ms between COTS units.

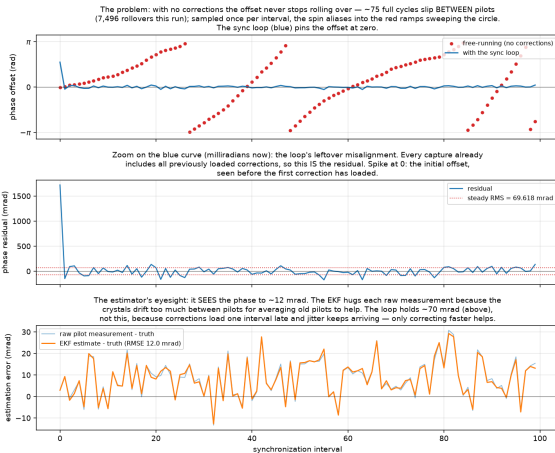
What failure looks like — and what the loop buys

The next page shows the whole problem and the whole fix in one figure (one-way loop, walkthrough form):

- **Top panel** — the raw phase difference between two free-running radios: it wraps through $\pm\pi$ roughly **7,500 times in 5 seconds** (red). With the closed loop on, the same physics holds flat at zero (blue).
- **Middle panel** — zoom on the held residual: steady RMS **69.6 mrad**, comfortably under the 314 mrad (18°) coherence line.
- **Bottom panel** — the estimator's eyesight: the EKF tracks the phase to **12 mrad**.

Takeaway: The estimator measures the phase to 12 milliradians, but the loop only holds it to 70 milliradians. Roughly a factor of six is lost between measurement and actuation. That gap is caused by correction cadence and latency rather than estimation quality, and it is the central research finding of this project.

One-way OTA phase synchronization (--model sdr)
 Method: master sends a known 4.6 ms Zadoff-Chu pilot every 50 ms → slave measures its phase & frequency → EKF smooths → slave retunes its own oscillator (correction loads 1 interval later). Tracking RMSE=12.017 mrad, steady residual RMS=69.618 mrad



The one-way synchronization walkthrough. The top panel shows the phase difference between two free-running radios, which wraps around roughly 7,500 times in five seconds (red); the blue trace shows the same radios with the control loop running, holding the phase flat. The middle panel zooms in on the held residual (69.6 milliradians RMS), and the bottom panel shows that the estimator itself tracks the phase to 12 milliradians.

The stakes, in meters

Converting sync quality into drone-detection range ($N = 6$ stations, 1 W each, -15 dBsm quadcopter, Swerling-1, $P_d \geq 0.9$ at $P_{fa} = 10^{-6}$):

	detection range	vs. single station
single station	757 m	1.00×
free-running (no sync)	1184 m	1.57×
published consensus (DFPC)	1972 m	2.61×
this work (star methods)	2896–2899 m	3.83×
perfect sync (bound)	2900 m	3.83×

Takeaway: Synchronization quality is worth ~ 1.7 km of detection radius; the methods built here operate within 0.1% of the physical bound.

Philosophy: no statistics-level shortcuts

Every claim is validated on **sampled IQ streams** through a full physical layer (Sionna, 3GPP TDL-D multipath), and every estimate is graded against an **oracle twin** of the same capture with noise removed — ground truth, never the receiver's own opinion of itself.

Signal chain (every capture)

clock/NCO → ZC frame → PA + 12-bit DAC → TDL-D multipath + shadowing + AWGN → slave LO + SFO + phase noise → IQ imbalance + AGC + 12-bit ADC → detect / time / coarse CFO / fine CFO / phase → iterated EKF → quantized NCO correction (arrives *late*).

Impairments always on (defaults)

3GPP TR 38.901 TDL-D, 100 ns delay spread; log-normal shadowing (2 dB, AR(1) 10 s); PA soft clipping; 12-bit DAC/ADC; AGC; IQ imbalance 0.2 dB/1°; DC offset; ±32-sample timing jitter; SFO tied to the same crystal as CFO; white-FM + white-PM + flicker-FM + frequency-walk oscillator noise; initial CFO 1.5 kHz; correction latency 1 interval; NCO quantization $2\pi/2^{16}$.

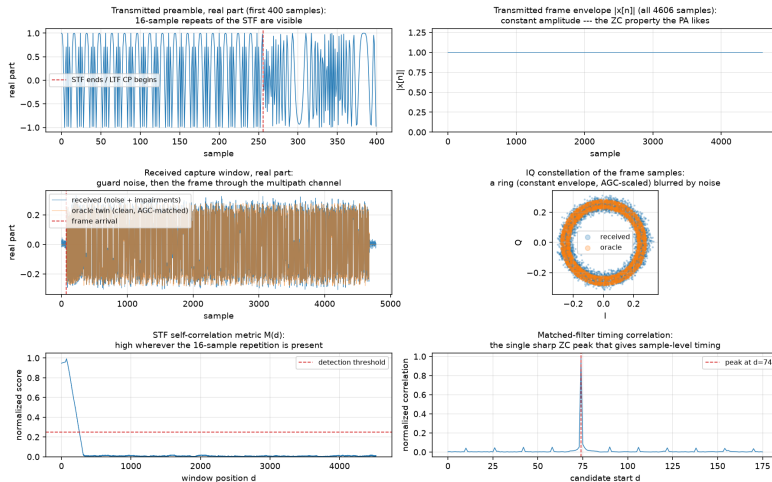
Deterministic: same seed ⇒ byte-identical run. 41 regression tests.

What a sync pilot actually is (not a tone)

The next page shows one capture at the raw-sample level. How to read it:

- **Transmitted frame:** Zadoff–Chu sequences — constant envelope (PA-friendly), 99% of power spread across ~ 800 kHz. A short training field (16-sample ZC $\times$ 16 repeats) for detection and coarse CFO, then two long cyclic-prefixed ZC-2047 fields for fine CFO and phase. 4.6 ms per capture at 1 MS/s.
- **Received frame:** the same waveform after multipath, noise, and hardware — next to its *oracle twin* (same channel, no noise), the grading reference.
- **Constellation:** a ring, not points — ZC is a polyphase sequence; impairments thicken the ring.
- **Correlations:** the STF self-correlation plateau (detection) and the single sharp matched-filter peak (sample timing through multipath) — the two things a tone cannot do.

One capture at the IQ level (TDL-D, 20 dB SNR, CFO 1500 Hz)



A single synchronization pilot, shown at the level of raw samples. The panels show the transmitted Zadoff–Chu frame, the signal that actually arrived after the multipath channel and hardware impairments, the received constellation, and the correlation peaks the receiver uses to detect the pilot and align its timing.

The estimator and the controller

Two-state iterated EKF per link, $x = [\theta, \omega]^\top$:

$$F = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix}, \quad x^- = Fx, \quad P^- = FPF^\top + Q, \quad z = [\cos m, \sin m, \omega_{\text{meas}}]^\top$$

- Cosine/sine embedding avoids phase-wrap discontinuities; 6 Gauss–Newton relinearizations; Joseph-form covariance.
- R from physics, not tuning: phase variance $\frac{1}{2\text{SNR}L} + \sigma_{pn}^2(\text{offset} + \text{span}/3) + \sigma_{wpm}^2/L$.
- Control with latency: correction = quantize($F^L x^+$) loads L intervals late (default 1). The controller forward-predicts through its own delay — what survives is its *frequency-estimate error* integrated over the horizon. This term is the novelty seed.

Takeaway: The loop is honest end-to-end: physical measurement noise, quantized actuation, late corrections.

The key equation: two-way reciprocity

Forward and reverse pilots cross the *same channel realization* within one interval:

$$m_{\text{fwd}} = \text{wrap}(\theta + \phi_c), \quad m_{\text{rev}} = \text{wrap}(-\theta + \phi_c)$$
$$\hat{\theta} = \text{wrap}\left(\frac{\text{wrap}(m_{\text{fwd}} - m_{\text{rev}})}{2} + b_{\text{chain}} - \hat{\omega} \tau_{\text{TDD}}/2\right)$$

The channel phase ϕ_c cancels — the *actual crystals* align, which is what open-loop coherent combining needs.

Two lessons discovered by simulation

(1) The half-difference carries a global π ambiguity; the 1-bit combining check that resolves it must be *periodic* — a coasting station can drift across the branch. (2) Never coast during acquisition: residual CFO makes one skipped interval fatal.

How to read every scoreboard in this deck

All sync methods are judged on ONE standard three-panel figure (identical axes, units, thresholds — so methods can be compared side by side):

- 1 **Top — the phase on the circle ($\pm\pi$):** red dots = free-running, rolling over constantly; blue = the loop holding it. The problem and the fix at a glance.
- 2 **Middle — |residual| in mrad, log axis,** against the red $314 \text{ mrad} = 18^\circ$ line (below it = 90%+ coherent gain). The yellow box prices the COST: the share of channel time spent on sync pilots.
- 3 **Bottom — 2-station coherent gain in %,** against the 90% threshold; red dots show the same gain with NO sync (sweeping 100%→0% as the drift passes through anti-phase).

Takeaway: Judge every method by accuracy AND cost together — e.g. micro beats two-way on residual (28 vs 84 mrad) but pays more airtime (26% vs 19%).

Method 1 — one-way: track what you can hear

How it works, one interval at a time:

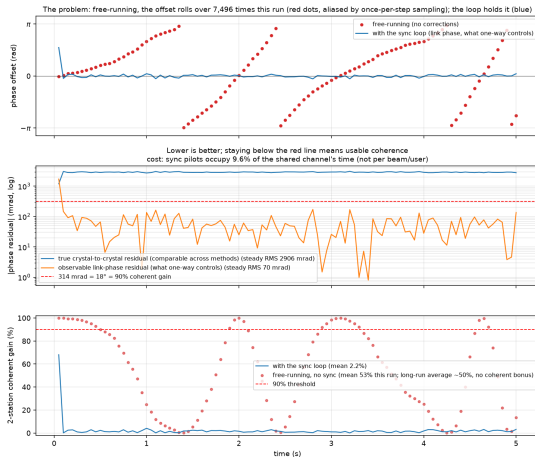
- 1 The master transmits the known ZC pilot frame.
- 2 The slave detects it, estimates timing, frequency offset, and phase from the matched filter — but every phase it measures is the *sum* $m = \theta + \phi_c$ (oscillators + channel).
- 3 The EKF tracks $[m, \omega]$ and the slave retunes its own NCO to drive $m \rightarrow 0$.

What that buys and what it costs (next page):

- The *observable* is held to **70 mrad** (orange) — the link looks perfectly locked.
- But θ and ϕ_c are not separately observable from a one-way pilot, so the loop simply absorbs the channel: the true crystal offset parks at $\theta = -\phi_c = \mathbf{2906\text{ mrad}}$ (blue), and the 2-station coherent gain is **2.2%** — nearly anti-phase while “locked.”

Takeaway: Cheapest possible loop (9.6% airtime), and fine for serving users with CSI — but for open-loop coherent combining the channel must be *cancelled*, not absorbed.

One-way OTA phase synchronization (—model sdr) — standard scoreboard
 Insight: one-way pins the radio LINK phase (orange) but the crystals park at the absorbed channel phase (blue, above the red line) — so open-loop coherent gain is lost. Two-way exists to fix exactly this.



The one-way method's scoreboard. The loop successfully holds the quantity it can measure (the oscillator phase plus the channel phase) to 70 milliradians, but the true oscillator-to-oscillator offset settles at 2906 milliradians, because the channel phase was folded into the correction. As a result the two stations combine at only 2.2 percent of their ideal coherent power, even though the link appears locked.

Method 2 — two-way: cancel the channel by symmetry

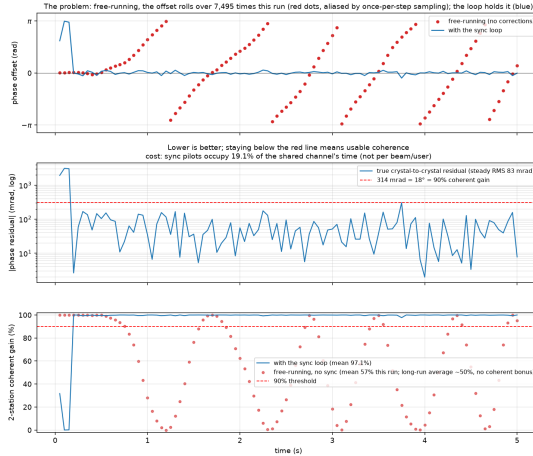
How it works, one interval at a time:

- 1 Station A transmits its pilot; B captures it and measures $m_{\text{fwd}} = \text{wrap}(\theta + \phi_c)$.
- 2 They swap roles (TDD, same frequency, ~ 1 ms turnaround); A measures $m_{\text{rev}} = \text{wrap}(-\theta + \phi_c)$ — the channel is *reciprocal*, so ϕ_c is the same in both directions.
- 3 Half the difference isolates the crystals: $\hat{\theta} = \text{wrap}(m_{\text{fwd}} - m_{\text{rev}})/2$ (+ chain-bias and turnaround-drift corrections). ϕ_c cancels exactly.
- 4 The EKF filters $\hat{\theta}$; the slave applies the forward-predicted, quantized correction one interval later.

Measured (next page): true crystal residual **83.5 mrad** RMS, coherent gain **99.83%** ($3.99\times$ of the $4\times$ ideal), at **19.1%** airtime (two 4.8 ms captures per 50 ms).

Takeaway: The actual crystals align, so the array can combine toward *any* target with known geometry — no per-target calibration. This is the workhorse every refinement builds on.

Reciprocal two-way synchronization (—model twoway)
 Method: both stations send each other a pilot over the SAME channel each interval; half the difference of the two measured phases cancels the channel, leaving the true crystal-to-crystal offset → EKF → slave retunes



The two-way method's scoreboard. Because the forward and reverse pilots cross the same channel, half the difference of the two measurements removes the channel phase entirely. The true oscillator offset settles at 83.5 milliradians, the two-station coherent gain reaches 99.83 percent of ideal, and the pilots occupy 19.1 percent of the channel.

Method 3 — micro: once locked, phase is cheap

The insight: the closed-loop residual is dominated by drift accumulated *between* corrections. Timing and wide-range CFO need the full 4.6 ms frame — but once the loop is locked, *phase alone* can be measured from a very short pilot.

How it works:

- 1 Run the full two-way exchange once per 50 ms interval (detection, timing, coarse/fine CFO, phase), exactly as in Method 2.
- 2 Between full frames, insert $M=4$ tiny reciprocal *micro-pilots*: one CP-protected ZC-255 sequence, 287 samples = **0.287 ms** (vs. 4.6 ms) — no detection preamble needed.
- 3 The receiver *rides the tracking state*: it derotates with its current frequency estimate and correlates at the expected arrival time. Corrections issue every sub-interval.

Measured (next page): **28.1 mrad** at **26.0%** airtime, gain 99.98%. The pilot-count knob: $M=0/2/4/9 \rightarrow 71.5/35.3/27.9/22.8$ mrad at 19.1/22.6/26.0/34.6% airtime.

Takeaway: $\sim 3\times$ tighter than two-way for +7 points of channel — accuracy bought with airtime.

Method 4 — hybrid: a smarter observer, cheaper pilots

The insight: one-way pilots are cheap and DO carry information — they just can't tell oscillator from channel. Give the filter a third state and let it *model* the entanglement:

$$x = [\theta \quad \omega \quad \phi_c]^\top \quad \text{one-way pilots observe } \theta + \phi_c; \quad \text{anchors observe } \theta \text{ alone.}$$

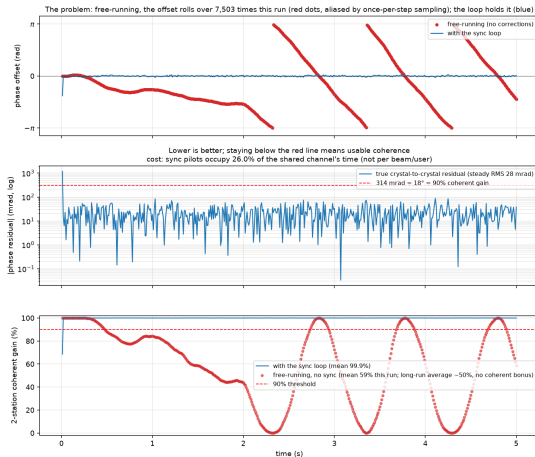
How it works:

- 1 Most intervals: cheap *one-way* micro-pilots track the SUM $\theta + \phi_c$ — enough to follow oscillator drift, since ϕ_c moves slowly (a static channel barely moves at all).
- 2 Every $K=5$ intervals: one full *two-way anchor* re-measures θ alone (channel cancelled), re-pinning the split between the two states.

Measured (next page): **33.8 mrad** at **14.9%** airtime, gain 99.97%. Anchor-cadence knob:
 $K=1/5/20 \rightarrow 31.8/32.9/34.0$ mrad — the cadence is set by *channel* coherence time, not oscillator quality (with Doppler at 0.2 m/s, $K=20$ collapses to 1752 mrad unless the channel prior is matched).

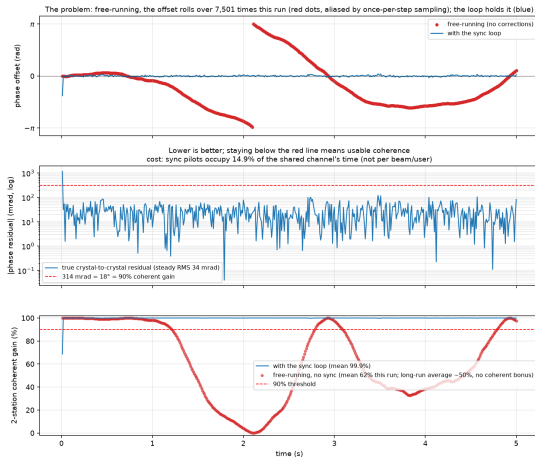
Takeaway: Most of two-way's accuracy at 3/4 of its cost — the accuracy-per-airtime winner, and (scaling section) the method that stretches the array-size wall furthest.

Two-tier micro-pilot synchronization (—model micro, 4 micro-pilots/interval)
 Method: one full two-way pilot exchange per interval (as in twoway), plus tiny reciprocal phase-only pilots in between — correcting several times per interval before drift can accumulate



The micro-pilot method's scoreboard. Four short phase-only pilots inserted between the full two-way exchanges correct the drift more often, bringing the residual down to 28.1 milliradians at the cost of 26.0 percent of the channel.

Hybrid calibration (—model hybrid, two-way anchor every 5 intervals)
 Method: cheap ONE-way pilots track oscillator+channel phase jointly; an occasional two-way exchange
 (the anchor) separates the two again — most of two-way's accuracy at a fraction of the airtime



The hybrid method's scoreboard. Inexpensive one-way pilots track the drift most of the time, and a full two-way exchange every fifth interval re-separates the oscillator phase from the channel phase. The residual is 33.8 milliradians at only 14.9 percent of the channel, the best accuracy per unit of airtime of any method tested.

Method 5 — consensus (DFPC): the published alternative

How it works (Rashid & Nanzer, IEEE TWC 2023): no master at all. Each node measures the other's pilot and retunes *itself* by half its estimated offset — both nodes move, and the pair meets in the middle:

$$\phi_a \leftarrow \phi_a - \frac{1}{2}\hat{\theta}, \quad \phi_b \leftarrow \phi_b + \frac{1}{2}\hat{\theta}.$$

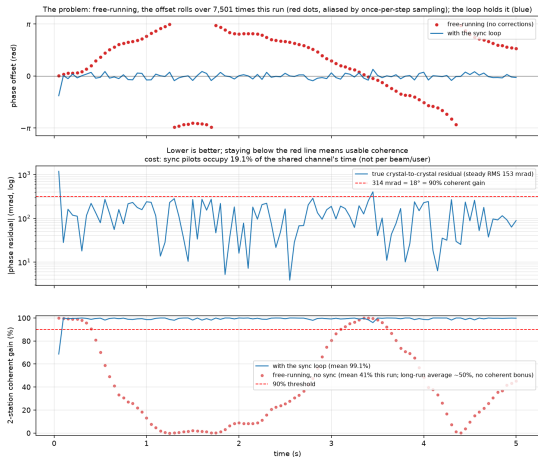
KF-DFPC adds a Kalman filter on the estimates. Reimplemented here over the full physical layer:

variant	residual	gain	note
naive, as published	3009 mrad	0.68%	anti-phase capture
DFPC + reciprocity	153.0 mrad	99.42%	next page
KF-DFPC + reciprocity	82.8 mrad	99.83%	page after

Why the published form breaks: the paper assumes channel-free measurements. Over a real channel each node's raw measurement contains ϕ_c , and the *wrapped* symmetric update then has TWO stable fixed points — in-phase and anti-phase. Both look “converged”; this seed lands on the wrong one (0.68% gain). Granting the paper its assumed side channel (“reciprocity steelman”) restores it; with filtering it reaches the same physical floor as our two-way.

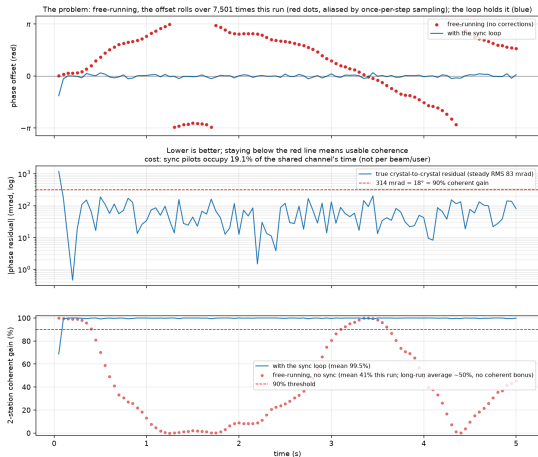
Takeaway: The idea is sound — but statistics-level validation, which never simulated a channel, could not see the second fixed point. Waveform-level validation is not optional.

dfpc consensus over the physical layer (--model dfpc)
 Method (Rashid & Nanzer 2023): each node measures the other's pilot and retunes its own oscillator by half its estimated offset each interval, so the two drift toward each other (no master/slave roles)



The scoreboard for DFPC when it is granted the channel-free measurement exchange its publication assumes. Under that assumption the consensus update works as intended, settling at 153 milliradians and 99.42 percent coherent gain.

kf-dfpc consensus over the physical layer (--model kfdfpc)
 Method (Rashid & Nanzer 2023): each node measures the other's pilot and retunes its own oscillator by half its estimated offset each interval, so the two drift toward each other (no master/slave roles)



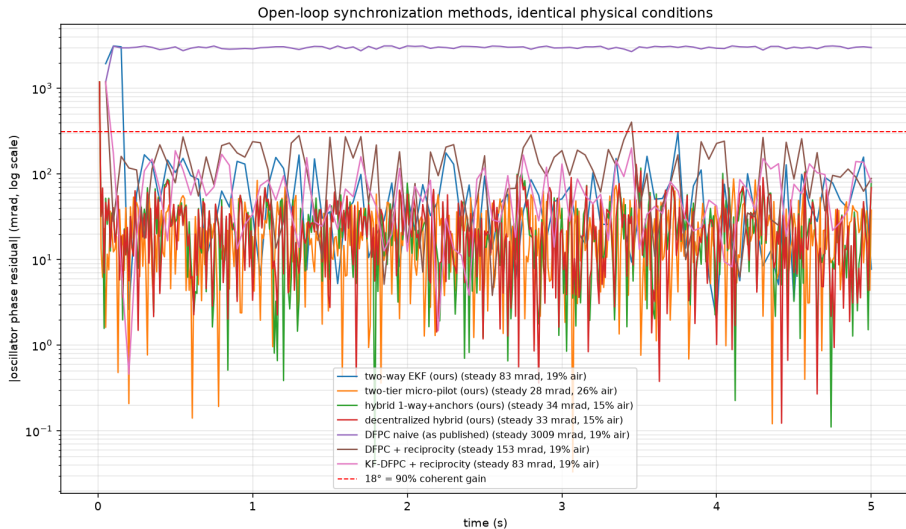
The scoreboard for KF-DFPC, the Kalman-filtered variant of the consensus algorithm, under the same assumption. The filtering brings it to 82.8 milliradians and 99.83 percent coherent gain, matching the physical floor that our two-way method reaches.

All methods, identical physics

The next page overlays every method's residual trajectory on one log axis (same seed, same channel, same impairments):

approach	residual (mrad)	coherent gain	airtime
two-tier micro-pilot (ours)	28.1	99.98%	26.0%
decentralized hybrid (ours)	33.4	99.97%	14.9%
hybrid 1-way+anchors (ours)	33.8	99.97%	14.9%
KF-DFPC + reciprocity	82.8	99.83%	19.1%
two-way EKF (ours)	83.5	99.83%	19.1%
DFPC + reciprocity	153.0	99.42%	19.1%
DFPC naive (as published)	3009.4	0.68%	19.1%

Takeaway: Every trajectory except naive DFPC settles below the 314 mrad line; naive DFPC is pinned at ~ 3000 mrad — the anti-phase fixed point, flat and stable, which is precisely why statistics-level validation never caught it.



All seven methods run under identical physical conditions, with their residual trajectories overlaid on one logarithmic axis. Every method except naive DFPC settles below the 314-milliradian coherence threshold. Naive DFPC sits flat near 3000 milliradians: it has converged, but onto the anti-phase solution.

The error floor: measured to 12 milliradians, held only to 70

The closed-loop residual decomposes into three terms:

$$\sigma_{\text{total}}^2 \approx \underbrace{\sigma_{pn}^2 f_s T}_{\text{drift: 45 mrad}} + \underbrace{(\sigma_{\omega} + L T)^2}_{\text{LATENCY: 38 mrad}} + \underbrace{\sigma_{\text{track}}^2}_{\text{estimation: 12 mrad}} \Rightarrow \sim 70 \text{ mrad (measured 69.6)}$$

ablation (each isolates one term)	sees (mrad)	holds (mrad)
full defaults	12.0	69.6
--no-rf-impairments (latency only)	1.1	34.3
--correction-latency 0 (tracking only)	12.3	12.3

Takeaway: The middle term — actuation latency coupled to the Kalman frequency posterior — appears in no published sync analysis (checked through Dec 2025). It is the dominant loss at defaults and the primary publishable claim.

Method 6 — decentralized hybrid: masterless, done right

How it works: keep hybrid's 3-state estimation (Method 4), but make the *control* symmetric like DFPC's — each node retunes half-way toward the other, so neither is the reference and the pair meets at its average clock.

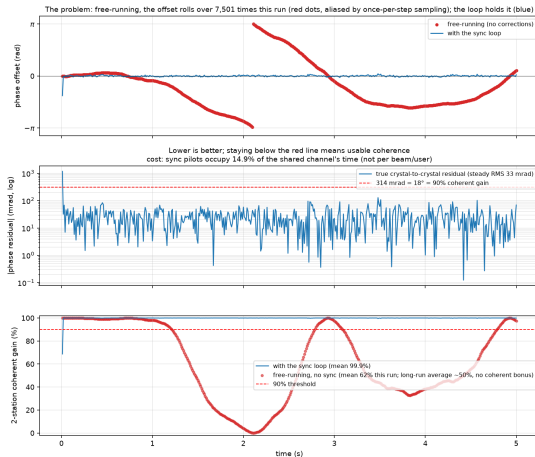
- **At $N=2$** (next page): 33.4 vs. 33.8 mrad — decentralizing the control is *free*. Gained: no single point of failure. Lost: a fixed datum (the common clock drifts freely).
- **At $N>2$** (page after): a true mesh — shared oscillators on a nearest-neighbor chain, degree-weighted corrections, and *periodic* 1-bit branch checks (a one-shot π -check is not enough: slow mesh convergence lets an edge drift to the wrong branch after an early check passes).

The *update schedule* then decides everything:

control law	structure	gain @ $N=4$	gain @ $N=6$
symmetric	DFPC-style simultaneous updates	74%	80%
alternating	edges take turns (masterless)	99.9%	95.8%
directed	elected-root tree	99.9%	99.8%

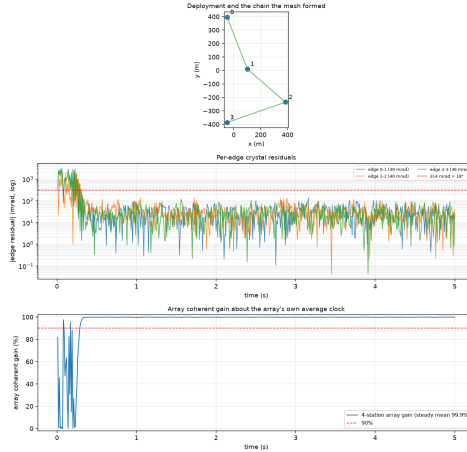
Takeaway: The 20–25-point cost is not decentralization — it is *simultaneous symmetric updates on shared nodes*. Taking turns recovers most of it; an elected root all of it, with a decentralized fault structure intact.

DECENTRALIZED hybrid calibration (--model dhybrid, anchor every 5 intervals)
 Method: identical estimation to hybrid (cheap one-way pilots + sparse two-way anchors), but NO master —
 each node retunes HALF-way toward the other (symmetric consensus control); the pair meets at its average clock



The decentralized hybrid pair. Each node retunes halfway toward the other, so there is no master station and no single point of failure. The accuracy, 33.4 milliradians, is statistically identical to the centralized version.

4-station DECENTRALIZED hybrid mesh (-model dhybrid --stations N)
 No reference: nearest-neighbor chain, symmetric degree-weighted corrections, periodic 1-bit branch checks.
 Interior edges pay a consensus tax (under-relaxed + disturbed by neighbors) — compare with the centralized star.



A true four-station decentralized mesh using alternating updates. The stations synchronize over a nearest-neighbor chain with no reference station. The per-edge residuals are 36 to 40 milliradians, and the array holds 99.91 percent coherent gain while using 44.7 percent of the channel.

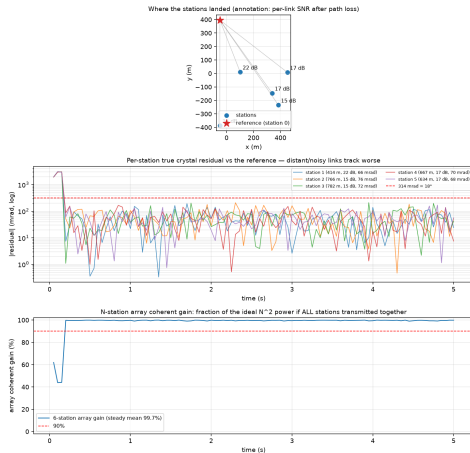
A real deployment: six stations, one shared channel

The network simulation places stations uniformly at random in a 500 m disc; each link's SNR follows log-distance path loss. Next page, two-way at $N = 6$:

station	distance	link SNR	steady residual	detect
1	414 m	22.2 dB	66.2 mrad	100%
2	766 m	15.0 dB	76.5 mrad	100%
3	782 m	14.8 dB	72.5 mrad	100%
4	667 m	16.6 dB	70.0 mrad	100%
5	634 m	17.2 dB	68.2 mrad	100%

Takeaway: Array coherent gain 99.65% — but the pilots already occupy **95.6%** of the shared channel at $N=6$. Accuracy is not the scaling problem; *airtime* is.

6-station synchronization (I-model two-way, star topology to the reference)
 random deployment in a 500 m disc, path-loss exponent 2.7, SNR 20 dB at 500 m
 total pilot airtime 96%



A six-station network sharing one channel. The map shows the random deployment with each link's signal-to-noise ratio; the other panels show each station's residual over time and the array's coherent gain, which averages 99.65 percent. The cost is that the synchronization pilots occupy 95.6 percent of the channel.

Two trades govern everything

Trade 1: cadence (next page)

Longer sync interval = more drift between corrections, fewer pilots:

interval	residual	airtime
10 ms	32.8 mrad	95.6%
50 ms	83.5 mrad	19.1%
200 ms	243.5 mrad	4.8%
500 ms	1035.9 mrad	1.9%

50 ms is one point on a curve, not a magic number.

Trade 2: array size (two pages on)

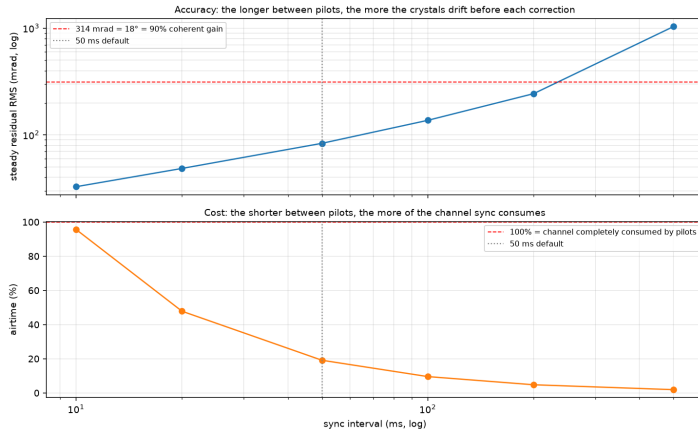
Pilots share ONE channel, so airtime grows linearly with N (hybrid shown):

N	worst residual	airtime
2	26.1 mrad	14.9%
4	30.7 mrad	44.7%
8	45.6 mrad	104.4%
12	45.6 mrad	164.1%

Accuracy survives scaling; the *channel* does not — the wall sits between $N=4$ and $N=8$.

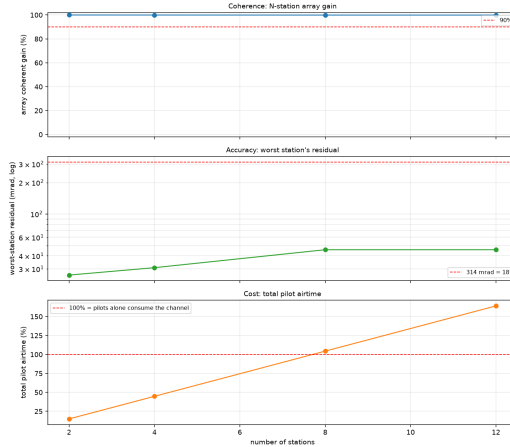
Takeaway: Cadence is the currency; the wall is a scheduling problem — which is where this project's frontier lies.

Choosing the pilot cadence (--model twoway)
The default 50 ms is one point on this trade curve, not a magic number: pick the longest interval whose accuracy still meets your target, because that costs the least airtime.



The cadence trade for the two-way method. Synchronizing more often holds the phase tighter but consumes more of the channel: a 10-millisecond interval achieves 33 milliradians but costs 95.6 percent of the channel, while a 500-millisecond interval costs only 1.9 percent but lets the residual grow past one radian. The 50-millisecond default is one point on this curve, not a special value.

Scalability (hybrid model): coherence and cost vs. array size
Bigger arrays promise N^2 combined power, but per-station errors compound (gain falls) and TDMA pilot
airtime grows linearly until sync alone consumes the channel — the two scaling walls.



The station-count trade for the hybrid method. Accuracy survives scaling, with the worst station staying under 46 milliradians at twelve stations. However, because all pilots share a single channel, the total pilot airtime grows linearly with the number of stations and exceeds the entire channel somewhere between four and eight stations.

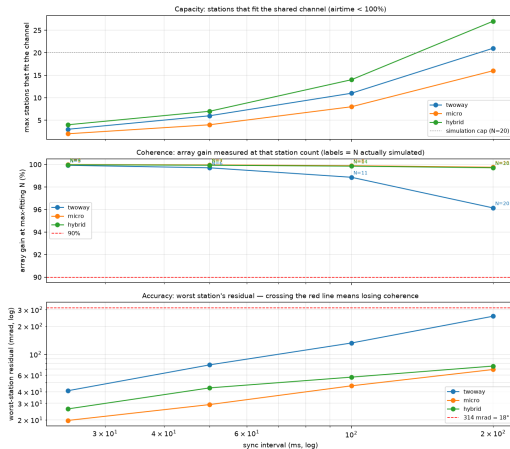
The full 2-D answer: which method scales best?

Sweeping cadence $\times$ array size together; for each method and cadence, the largest N that fits the channel and the gain held there (next page). Measured verdict:

cadence	two-way	micro	hybrid
25 ms	$N=3$	$N=2$	$N=4$
50 ms	$N=6$	$N=4$	$N=7$
100 ms	$N=11$	$N=8$	$N=14$
200 ms	$N=21+$ (96.1%, worst 256 mrad)	$N=16$	$N=27+$ (99.7%, worst 75 mrad)

Takeaway: Hybrid dominates the whole grid: at 200 ms it holds 20 stations at 99.7% gain on 71% of the channel, while two-way's worst station has degraded to 256 mrad and micro fits only 16. Cheap pilots + a smarter observer = the furthest wall.

Which method scales best? cadence $\times$ station count, per method
 Longer intervals fit more stations (pilots get cheaper) but let crystals drift more between corrections.
 The winner holds the most stations while staying accurate — read all three panels together.



The combined answer across cadence and array size. For each method and each synchronization interval, the figure reports the largest array that fits in the channel and the coherent gain the array holds at that size. The hybrid method supports the largest arrays at every cadence tested.

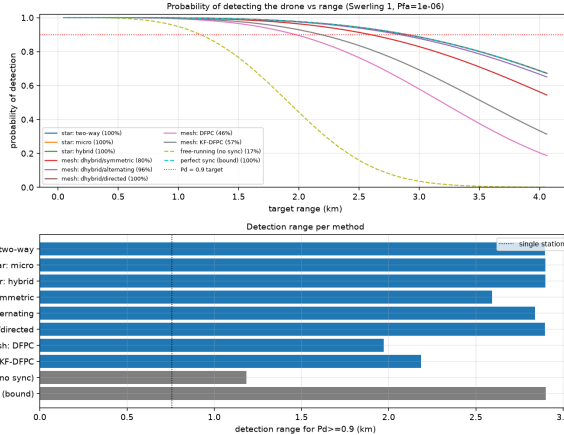
From sync gain to detection range (analytic layer)

$\text{SNR}_{\text{det}} = N^3 G^2$ with each method's *measured* gain G , Swerling-1 closed forms (next page):

method	sync gain	detect range	vs. perfect
star: two-way / micro / hybrid	99.7–99.9%	2896–2899 m	99.9–100%
mesh: dhybrid directed	99.8%	2897 m	99.9%
mesh: dhybrid alternating	95.8%	2839 m	97.9%
mesh: dhybrid symmetric	80.0%	2594 m	89.4%
mesh: KF-DFPC	56.7%	2185 m	75.3%
mesh: DFPC	46.2%	1972 m	68.0%
free-running	16.7%	1184 m	40.8%
perfect sync (bound)	100%	2900 m	100%

Takeaway: Every sync result converts directly into operational range: the consensus-tax costs 245–305 m; the published DFPC gives up ~ 930 m; free-running loses the coherent prize entirely.

Drone-detection viability of each synchronization method (N=6, RCS -15 dBsm, 1 W/station)
 Detection SNR = $N^3 \cdot G^2$ x single station, where G is each method's MEASURED array coherent gain —
 sync errors cost detection range twice (transmit focusing and receive combining).



Detection performance predicted from each method's measured synchronization quality. The curves show the probability of detecting the drone as a function of range, and the bars show the resulting detection range. The well-synchronized methods reach 2896 to 2899 meters of the 2900-meter perfect-synchronization bound, while the unsynchronized array reaches only 1184 meters.

The 3-D scene behind the waveform test (Sionna RT)

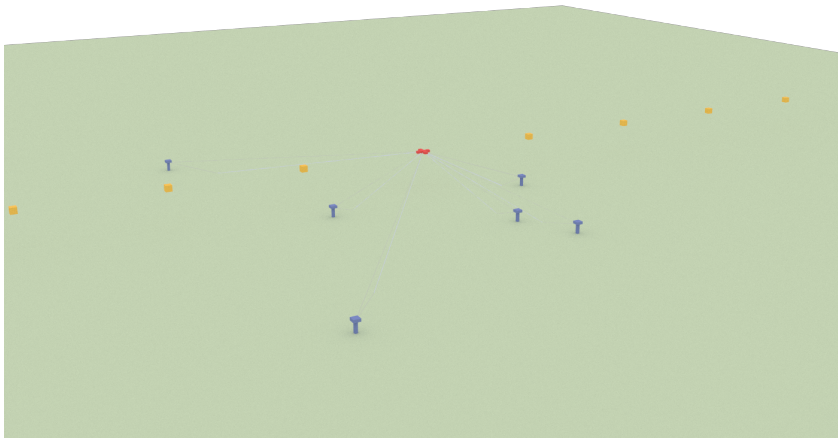
The detection Monte Carlo does not use closed-form propagation: station→target legs are **ray-traced** in a 3-D scene (next two pages):

- Base-station masts at 15 m, drone at 60 m, dielectric ground plane (medium-dry, $\epsilon_r = 15$) — the two-ray ground bounce that dominates low-altitude UHF is in every measured P_d .
- The drone itself is a *point probe + analytic RCS*: shooting-ray solvers physically cannot sample bounces off a 0.3 m object at hundreds of meters (verified: 0 paths). This RT-legs + analytic-RCS coupling is the standard ISAC approach.
- Validation: RT in free space reproduces the analytic model waypoint-for-waypoint to float32 precision.

Takeaway: First page: the deployment — masts (blue), flight path (orange), drone (red, enlarged for visibility). Second page: the solver-traced rays — line-of-sight plus one ground reflection from every station converging on the drone.



The three-dimensional scene used for ray-traced propagation. The six base-station masts are shown in blue, the drone's 2.4-kilometer flight path in orange, and the drone itself in red, all above a dielectric ground plane. The drone and waypoint markers are drawn larger than life so they are visible at this scale.



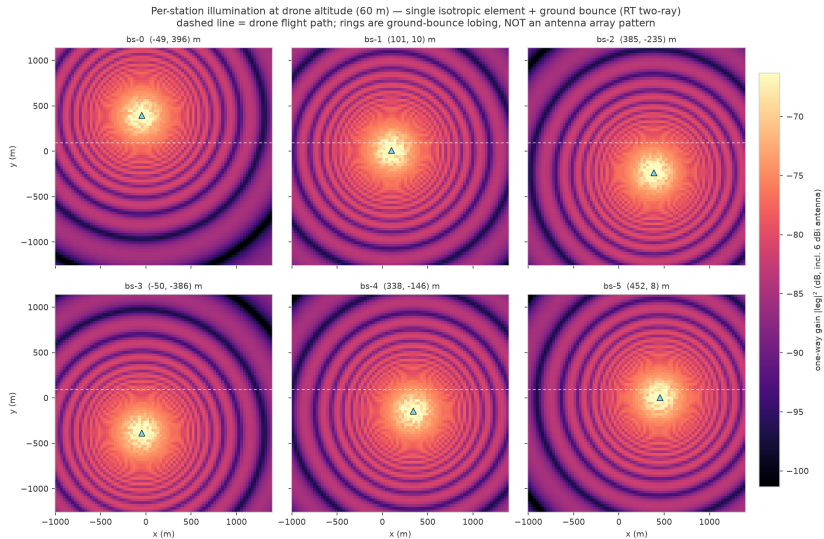
The propagation paths found by the ray-tracing solver. A direct line-of-sight path and one ground reflection from every station converge on the drone. These paths, including the ground bounce, are what feed the detection simulation.

What each station actually illuminates

Each station has a **single isotropic element** — no panel, no local beam. Its real “radiation pattern” at drone altitude is propagation physics (next page):

- Concentric bright/dark rings around each mast: two-ray interference between the direct ray and the ground bounce at the 60 m plane.
- A drone crossing the rings sees per-station illumination swing by tens of dB — this is what the RT layer adds to the measured P_d and why free-space link budgets mislead at low altitude.

Takeaway: The rings are ground-bounce lobing, NOT an antenna pattern — the array’s directivity lives in the phase relations *between* stations, shown on the following two pages.



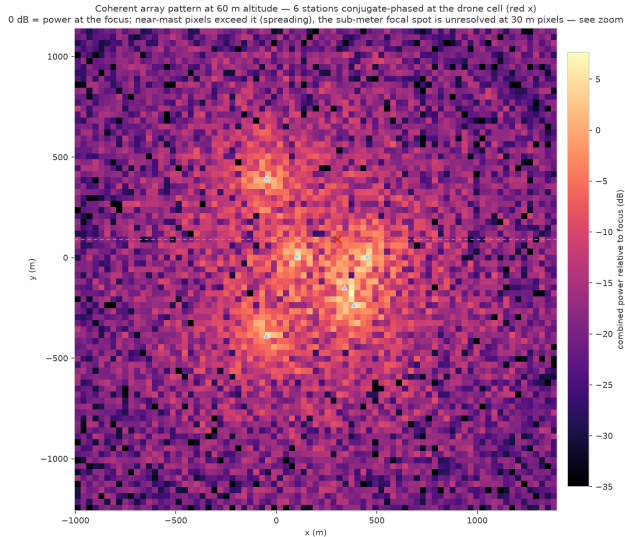
What each station actually illuminates at the drone's altitude of 60 meters. Each station has a single omnidirectional antenna, so the ring patterns are not antenna beams; they are interference between the direct ray and its reflection off the ground. A drone crossing these rings sees the illumination from each station swing by tens of decibels.

The array's “beam”: a sub-meter focal spot

Conjugate-phasing all six stations at one hypothesized cell is *near-field focusing*, not far-field beamforming. Next two pages:

- **Wide map** (30 m pixels): no beam shape at deployment scale — speckle brightening toward the array; near-mast pixels exceed the focal level (spherical spreading beats focusing up close; 0 dB $\equiv$ power at the focus).
- **Zoom** (2.5 cm pixels): the actual focal spot — ~ 25 cm wide ($\lambda R/D$ at 33 cm carrier over a ~ 1 km aperture), 0 dB at the focus, in a lattice of quasi-lobes with median -8.2 dB (the $1/N = -7.8$ dB sparse-array speckle floor).

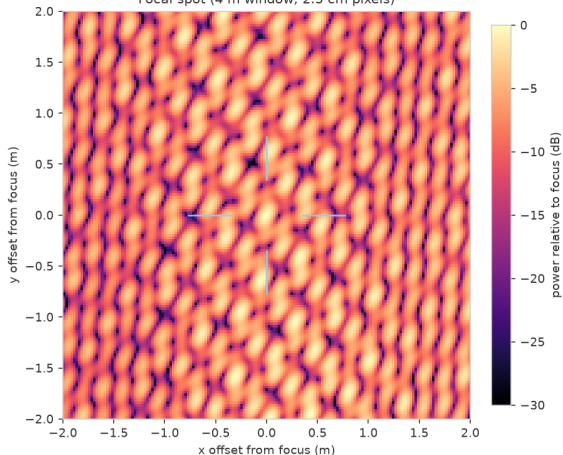
Takeaway: Six elements buy a razor-thin focal gain surrounded by high sidelobes — exactly why the detection test is *cued* (one cell hypothesis at a time, steered computationally).



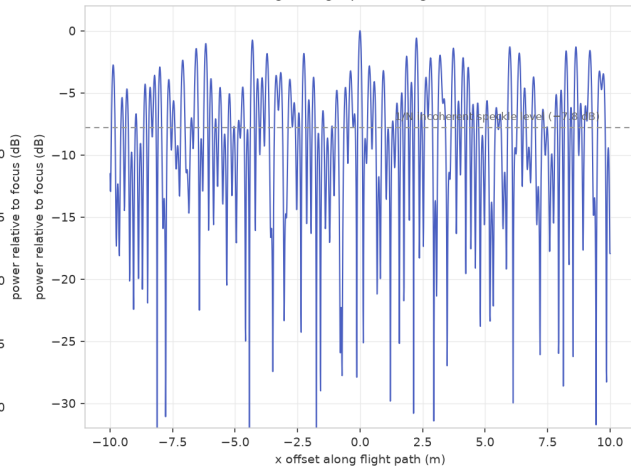
The combined power of all six stations, phased to focus on the drone's position (marked with the red cross), computed over the entire deployment. At this scale there is no beam; the pattern is speckle. The zero-decibel reference is the power delivered at the focus, and the pixels near the masts exceed it simply because they are much closer to a transmitter.

Distributed-array focal region — 6 elements over ~1 km aperture, 915 MHz, RT ground-bounce legs, perfect sync

Focal spot (4 m window, 2.5 cm pixels)



Cut along the flight path through the focus



A close-up of the focal region. The array concentrates its power into a spot roughly 25 centimeters wide, surrounded by sidelobe speckle whose median is 8.2 decibels below the focus, matching the one-over-N level expected for a six-element sparse array. This is why the detection test examines one hypothesized cell at a time.

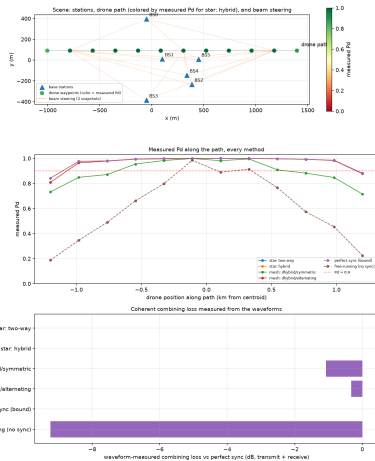
Counted detections along a drone crossing

The real test: no detection formulas anywhere in the loop. Every station transmits a 5G-style OFDM burst pre-steered at the hypothesized cell; echoes are synthesized sample-by-sample (RT legs, Swerling-1 RCS, *measured* sync residuals, real noise), matched-filtered, coherently combined, compared to an **empirical** threshold calibrated on target-absent trials. 1500 trials per waypoint; measured $P_{fa} = 1.00\text{e-}3$ (target 0.001). Next page: the scene, P_d along the path, and waveform-measured combining loss:

method	combining loss	P_d along path
star: two-way	-0.00 dB	84 98 98 99 100 ... 100 99 98 88 %
star: hybrid	+0.01 dB	identical
perfect sync (bound)	-0.00 dB	identical
mesh: alternating	-0.33 dB	81 96 98 99 100 ... 99 98 88 %
mesh: symmetric	-1.08 dB	73 85 87 95 98 ... 88 85 71 %
free-running	-9.24 dB	19 35 49 66 80 ... 57 45 22 %

Takeaway: The synced array is indistinguishable from perfect sync, waypoint for waypoint, in counted detections.

Waveform-level drone detection along a flight path (N=6, OFDM burst, RCS -15 dBsm, propagation: rt-ground)
 ZC pilots synchronize the stations; a 5G-style OFDM burst does the sensing; station-target legs are ray-traced
 (Sonnet RT, ground bounce included). Pd is counted from real sample streams with the measured sync residuals.



The waveform-level detection study. The map shows the deployment and the drone's path, colored by the measured probability of detection at each waypoint, with snapshots of the array refocusing as the drone moves. The lower panels give the detection probability along the path for each synchronization method, and the combining loss that each method's measured phase errors caused.

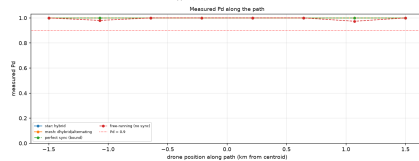
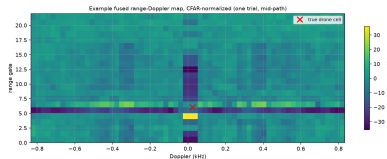
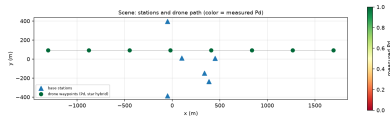
Field realism: clutter-limited detection

The realism layer adds what bites in the field, all injected into the raw sample streams (next page):

- Gamma-distributed ground clutter with internal motion — peak clutter-to-noise **70.1 dB**: the problem is clutter-limited, exactly like real UHF drone radar.
- Direct-path self-interference at true delays, removed by least-squares cancellation: **46.1 dB** $\rightarrow$ **6.6 dB** above noise.
- Aspect-dependent body RCS + rotor-blade micro-Doppler; a 64-pulse CPI with range–Doppler processing; CA-CFAR with the clutter ridge notched, calibrated on target-absent trials *containing* clutter.

Takeaway: Detection survives by *Doppler separation* — the physically correct mechanism. Honest note: at this geometry even the unsynced array detects ($\geq 97\%$) inside ~ 1 km — sync earns its keep at the coverage edge, which is exactly where the scheduling work aims it.

Clutter-limited drone detection (N=6, 64-pulse CPI, gamma -15 dB, micro-Doppler target)
 Ground clutter + direct-path interference + CFAR search over the range-Doppler map; Pd counted from real streams.
 Peak CNR 70 dB — detection survives by Doppler separation, not link budget.



The clutter-limited detection study. The center panel shows a processed range–Doppler map: the bright ridge at zero Doppler is ground clutter, which the detector notches out, and the drone appears as the marked cell offset in Doppler by its motion. The bottom panel shows the counted detection probability along the path.

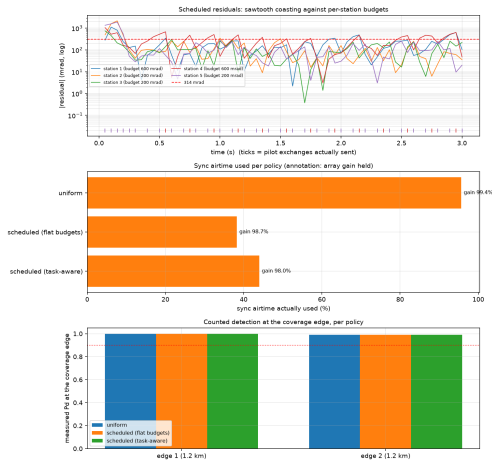
Idea: stop polling, start scheduling

Every published scheme polls every station at a fixed cadence. But each link's Kalman filter already *knows* its posterior uncertainty — so service a station only when its predicted uncertainty approaches its coherence budget, and set budgets by *detection utility* (tight for stations that matter at the coverage edge). Measured ($N=6$, next page):

policy	airtime	array gain	edge P_d
uniform	95.6%	99.4%	99.8 / 99.1%
scheduled (flat budgets)	38.2%	98.7%	99.8 / 99.1%
scheduled (task-aware)	44.0%	98.0%	99.8 / 99.1%

Takeaway: $\sim 55\text{--}60\%$ of the channel returned at *identical counted detection*. The figure shows the mechanism: sawtooth coasting against per-station budgets; tick marks are the pilots actually sent — the whitespace between them is the dividend.

Smart allocation of phase sync (N=6 star): service a station only when its filter says so
 Budgets are task-aware (tight for stations that matter at the detection edge). Ticks in the top panel are the pilots actually sent — the whitespace between them is the airtime returned to sensing.



Uncertainty-driven pilot scheduling. The top panel shows each station's residual sawing up and down between corrections, with tick marks at the pilots actually transmitted; stations are left to coast until their predicted uncertainty approaches their budget. The lower panels show the channel time used by each policy and the counted detection performance at the coverage edge, which is unchanged.

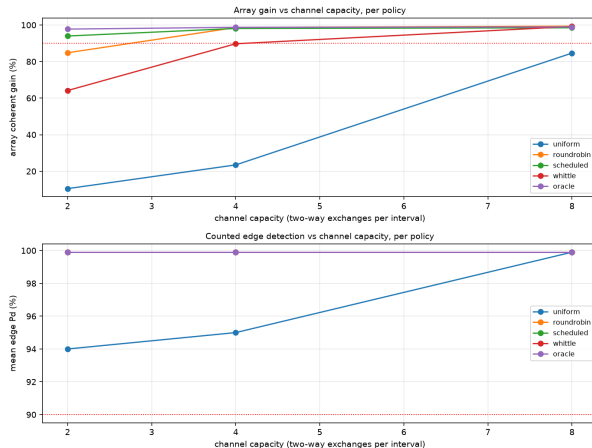
Under contention, scheduling decides who stays coherent

Cap the channel below demand ($N=10$: 9 links want service, only 2 exchanges/interval fit) and compare five policies — including a genie *oracle* that ranks by the true residual no online policy can see (next page):

policy	array gain	worst-station RMS
uniform	10.5%	(starved: never acquires)
round-robin	84.8%	1103 mrad
scheduled (ours)	94.0%	531 mrad
genie oracle	97.8%	218 mrad

Takeaway: Fixed cadence starves stations by construction; the Kalman-threshold rule captures most of what perfect information could buy. Honest negative result: a myopic Whittle-style index *underperforms* (64%) — it chases already-blown links instead of protecting salvageable ones.

Scheduling under contention (N=10 star, demand 9 exchanges/interval)
When the channel cannot carry a fixed cadence, informed scheduling decides which stations stay coherent.



Scheduling under contention, where nine links compete for a channel that can carry only a fraction of them each interval. The panels show array gain and edge detection as a function of channel capacity for each policy. The fixed uniform schedule permanently starves some stations and collapses, while the uncertainty-driven scheduler keeps the array coherent.

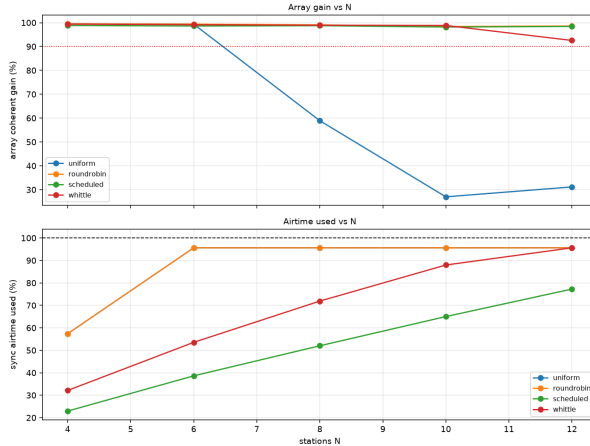
The wall, measurably moved

Pin the channel at its physical limit (5 exchanges/interval) and sweep N (next page):

policy	$N=6$	$N=8$	$N=10$	$N=12$
uniform	99.4%	58.9%	27.0%	31.2%
round-robin	99.4%	99.1%	98.4%	98.6%
scheduled	98.6%	98.7%	98.2%	98.4%
(scheduled airtime)	(38.6%)	(52.0%)	(65.0%)	(77.3%)

Takeaway: $N_{\max}$ holding $\geq 90\%$ gain: uniform = 6 — exactly the $19.1\% \times (N-1)$ arithmetic — and it collapses beyond. Scheduled holds 98% at $N=12$ using 77% of the channel, wall beyond the sweep: “the airtime wall moves right” is now a measured curve, not a sentence.

The airtime wall, per scheduling policy: gain and channel cost vs array size
(channel capacity pinned at its physical limit; fixed cadence saturates it, scheduling spends only what the filters demand)



The array-size wall for each scheduling policy, with the channel fixed at its physical capacity. The uniform schedule holds until exactly six stations and then collapses. The uncertainty-driven scheduler holds 98 percent coherent gain through at least twelve stations while using 77 percent of the channel.

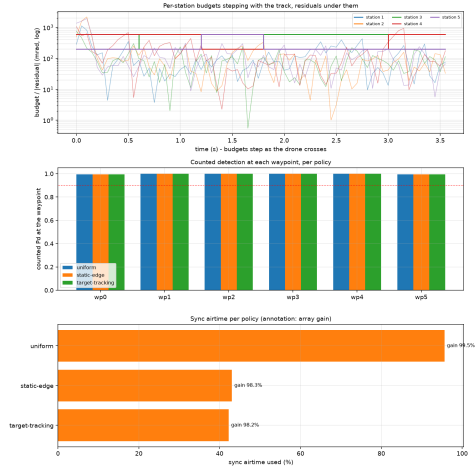
Closing the loop: budgets that follow the target

A drone crosses the deployment; each track segment re-targets the per-station budgets from the *same ray-traced legs the detector uses* — tight budgets for the stations that matter at the current target hypothesis, coast for the rest (next page):

- **56% of airtime returned** vs. uniform (95.6% $\rightarrow$ 42.2%), gain 98.2%.
- Counted P_d at every waypoint, from that segment's own residuals: **matching** (99.5–100% everywhere).
- Sensing $\rightarrow$ sync allocation $\rightarrow$ sensing: the closed ISAC loop, mechanically working.

Takeaway: At an uncontended operating point, tracking budgets tie static ones — the differentiating experiment (tracking budgets + *capped channel*) is built and queued.

Sensing-in-the-loop pilot scheduling (N=6): budgets follow the drone across the deployment
Each segment re-targets the budgets from the ray-traced legs toward the CURRENT target hypothesis.



Scheduling with budgets that follow the sensing target. As the drone crosses the deployment, each segment of its track re-assigns tight phase budgets to the stations that matter most for the current target position. The panels show the budgets stepping over time with the residuals beneath them, the counted detection at each waypoint, and the channel time used by each policy.

Two more dividends: hardware diversity and cheap pilots

Heterogeneous fleets (OCXO + TCXO mix)

The scheduler reads each link's posterior: OCXO stations serviced 12% of intervals, TCXO stations 55% — 97.5% gain at 36% airtime. The dividend *grows* with fleet spread, exactly as the coasting-time rule predicts.

Motion stress (candidate new physics term)

Scheduled coasting degrades with channel Doppler (145→227 mrad at 3 m/s) while uniform holds — the coast rule is missing a *channel-decorrelation* term. Same discovery shape as the latency term.

Multi-fidelity servicing

Posterior-driven choice per service: full two-way frame, or — once frequency is well-known — a 287-sample phase-only micro-pilot at its true sample cost:

airtime 57.4% → 10.4%

at unchanged gain (162 of 180 services went micro).

Engineering guarantee

All extensions opt-in; defaults reproduce the original scheduler *bit-for-bit* (regression-locked); 41 tests.

Where the novelty actually is

- 1 **Latency-coupled error floor** (confirmed unpublished through Dec 2025): the $(\sigma_{\omega} + LT)^2$ term — actuation latency as a design parameter distinct from cadence. Kill risk to clear: classical delayed-PLL/ADPLL literature.
- 2 **Sync airtime as an allocated resource** (built + measured here): uncertainty-driven, task-aware pilot scheduling, validated by counted detection through contention and the wall. Kill risk to clear: Age-of-Information / remote-estimation scheduling — must be framed as the intersection (physical pilots, coherence budgets, detection utility).
- 3 **What statistics-level validation hid**: anti-phase capture of published DFPC over a real channel; the consensus tax as an update-schedule artifact (74%→99.9% by taking turns). Bridging results, not standalone papers.

Takeaway: The detection layer itself is supporting material (established field since ~2009) — its role is converting milliradians into meters and P_d .

Next steps

On the runway

- Error-floor letter (IEEE WCL/LMWT class)
- Scheduling paper: run tracking budgets *under contention*; Aol prior-art pass first
- **Posterior-gated membership**: a station past $\sim 90^\circ$ *subtracts* from the beam — bench it until re-synced; soft weight $E[e^{j\theta}] = e^{-\sigma^2/2}$ from the filter. Joint airtime + membership scheduling appears unclaimed.
- Derive the Doppler coast-decorrelation term
- Two-SDR hardware point to anchor the simulation

Long-range (speculative, high ceiling)

- *Coherence capacity*: the minimum channel resource any protocol needs to hold N oscillators within budget — a law, connecting to the thermodynamic cost of timekeeping
- Synchronization phase transitions over real multipath at large N (statistical mechanics of engineered sync)
- Infrastructure-as-instrument: a cellular grid as one coherent aperture (hardware program)

Honest limitations

- **Simulation only** — no hardware validation anywhere in the stack.
- Statistical TDL channel (RT layer adds ground bounce + simple boxes, no terrain); flicker FM is an AR(1) surrogate; AGC instantaneous; two-way side channel error-free.
- N -station stars simulate links independently (reference noise per-link $\Rightarrow$ reported gains slightly conservative); broadcast amortization overstates star airtime.
- Detection: cued single gate, no tracker, no discrete clutter, PA not applied to the sensing burst; sparse-array speckle $\sim 1/N$ is why detection is cued.
- Hybrid's one-way frequency observation biased by LOS Doppler (4th state planned).

Takeaway: Every limitation above is documented in the repository next to the code it applies to — the claims are scoped to what was actually simulated.

Reproduce everything

Every figure in this deck regenerates from the repository (.venv/bin/python, seed 0):

```
python -m pytest                                # 41 tests, ~10 s
python simulation.py --plot-story                 # the problem + the loop
python simulation.py --plot-iq                   # what a pilot is
python simulation.py --model twoway --plot        # crystal alignment
python simulation.py --model compare --plot      # all methods head-to-head
python simulation.py --model twoway --stations 6 --plot # deployment
python simulation.py --model hybrid --sweep-stations --plot # the wall
python simulation.py --model twoway --sweep-interval --plot # cadence
python scalability_sweep.py                      # the full 2-D grid
python detection_study.py                        # ranges (analytic)
python waveform_detection_study.py               # counted Pd (RT)
python realistic_detection_study.py              # clutter-limited
python smart_sync_study.py                      # scheduling
python contention_study.py                      # contended channel
python airtime_wall_study.py                   # the wall, moved
python sensing_loop_study.py                   # budgets follow target
python render_scene.py ; python radiation_maps.py # 3-D scene + patterns
```

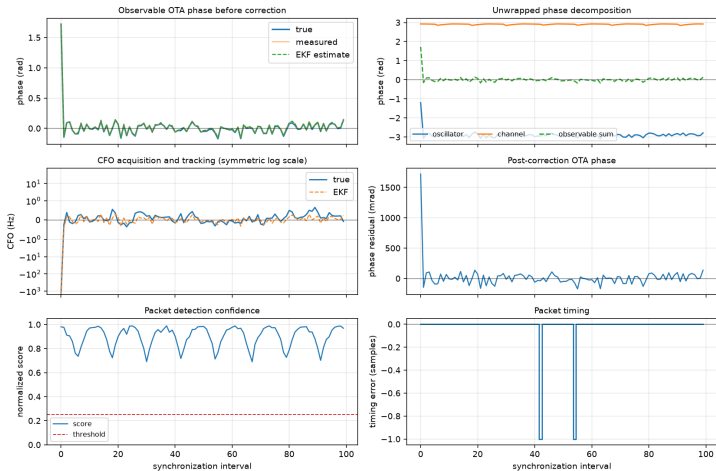
Docs: [HOW_TO_RUN.txt](#) · [METHOD_CALCULATIONS.md](#) · [CODE_REFERENCE.md](#) · [PUBLISHABLE.md](#) · [RESEARCH_IDEAS.md](#)

Backup: extra material

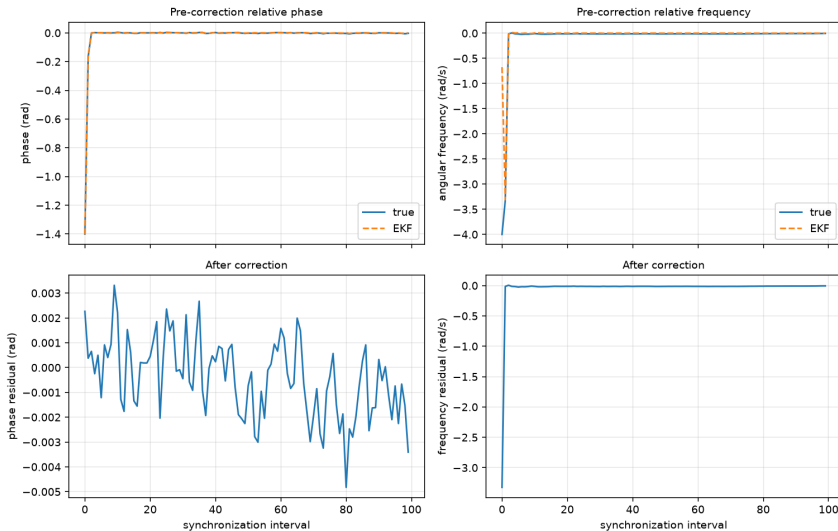
Two more full-page figures follow:

- **Receiver diagnostics** — the six-panel debugging view behind every scoreboard number: detection confidence, coarse/fine CFO acquisition, sample timing, and the phase decomposition.
- **The toy baseline** — the original tone-plus-noise model (no channel, no impairments; phase RMSE 1.6 mrad): what sync research looks like *before* the physical layer is allowed to object, kept only for comparison.

SDR OTA synchronization diagnostics — one-way method: known pilot → detect → measure phase/CFO → EKF → retune
detection=100.0%, OTA phase RMSE=12.017 mrad, CFO RMSE=0.196 Hz



The receiver's diagnostic view for a one-way run: detection confidence, coarse and fine carrier-frequency acquisition, sample-timing alignment, and the decomposition of the measured phase into its oscillator and channel components.



The original idealized model, with no channel and no hardware impairments, kept only for comparison. It achieves a residual of 1.6 milliradians. The difference between this and the 70 milliradians of the realistic loop is the price of the physical layer.

Summary

- ❶ **Problem:** distributed radios only combine coherently if their crystals agree to $\sim 18^\circ$ at 915 MHz — worth N^3 in detection SNR, ~ 1.7 km of range at $N = 6$.
- ❷ **Method:** an impairment-complete, waveform-level, oracle-graded testbed; six sync methods on one scoreboard.
- ❸ **Results:** two-way/hybrid/micro hold 28–84 mrad (99.8–100% gain); the closed-loop floor decomposes with a previously unpublished latency term (sees 12, holds 70); published consensus fails over a real channel in ways statistics-level validation could not see.
- ❹ **Frontier:** synchronization as a *scheduled resource* — half the channel returned at identical counted detection, the airtime wall measurably moved, and a closed sensing→sync loop running.

Thank you — deck + code + docs are one repository; every number regenerates with one command.